TRIGA Mark I

Independent neutronic and thermal-hydraulic reference calculations in GeN-Foam and Cardinal

A code-to-code study of a 250 kW pool-type research reactor, comparing a deterministic OpenFOAM multiphysics solution (GeN-Foam) against a stochastic Monte Carlo reference (Cardinal).

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GeN-Foam	2-group diffusion, finite volume
Cardinal	OpenMC / MOOSE / NekRS — continuous-energy Monte Carlo
Nuclear data	ENDF/B-VIII.0 (VII.1 for the Cardinal coupled run)
Repository	`github.com/kavyawadhwa134/triga-crosscode-verification`
Blind comparison	Cardinal frozen by git tag before GeN-Foam results were read

STATUS OF THIS DOCUMENT

This report presents **verification and a cross-code comparison**. It is **not a validation report**. No experimental data was used in either calculation, and agreement between two codes is not a substitute for measurement. Section 5 states the status of each verification level explicitly.

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1 Problem definition

1.1 Objective

Two separate calculations of the same TRIGA Mark I configuration were performed using fundamentally different numerical methods, with the goal of establishing a cross-code reference. Only the scientific problem definition was shared between the two models. The comparison was made blind: the Cardinal results were frozen under the git tag `pcm-independent-v1` before any GeN-Foam result was read, which is verifiable from the repository history.

The TRIGA Mark I is a pool-type research reactor whose defining characteristic is uranium–zirconium hydride fuel. Hydrogen bound in the ZrH lattice produces a large, prompt, negative fuel temperature coefficient, which is the physical basis of the reactor's inherent safety. Reproducing that feedback correctly is the principal physics requirement on any TRIGA model.

1.2 Specification

Table 1. Shared problem specification. Values marked *assumed* were not given and are documented modelling choices.

Parameter	Value	Basis
Fuel outer radius	1.815 cm	specification
Clad outer radius	1.866 cm	specification
Lattice pitch (square)	4.235 cm	specification
Active height	60.0 cm	specification
Core	0.495 m × 0.60 m	single zone, vacuum; shape not stated
Fuel U–$ZrH_{1.65}$	6.0 g cm^{-3}	U-235 1.7, U-238 6.8, Zr 89.862, H-1 1.638 wt%
Clad SS304	8.00 g cm^{-3}	density *assumed*
Coolant	1.0 g cm^{-3}	H_2O, unborated
Thermal power	250 kW	specification
Coolant inlet	293 K, 0.2 m s^{-1}	specification
Pressure	0.1 MPa	*assumed* — open pool
Energy groups	2, cutoff 0.625 eV	specification

The core extent "0.495 × 0.60 m" admits more than one reading. Cardinal models it as a 3-D cylinder of diameter 0.495 m; GeN-Foam models it as a 2-D slab 0.495 m wide and 0.60 m tall, infinite in the third direction. This difference in interpretation turns out to dominate the core eigenvalue comparison (Section 4).

There is no fuel–clad gap: the specification gives two radii, so the clad inner radius equals the fuel outer radius. Thermal scattering $S(\alpha, \beta)$ data for `c_H_in_ZrH` and `c_Zr_in_ZrH` is mandatory; without it the bound-hydrogen physics that defines a TRIGA is absent.

1.3 The two approaches

Table 2. The two calculations solve different equations. This distinction governs the interpretation of every comparison in Section 4.

	GeN-Foam	**Cardinal**
Neutron transport	Deterministic, 2-group diffusion	Monte Carlo, continuous energy
Energy treatment	2 groups, 0.625 eV cutoff	Continuous, full spectrum
Core geometry	2-D slab, homogenised	3-D cylinder, heterogeneous (+ homogenised)
Thermal–fluid	Finite volume, OpenFOAM	Spectral element, implicit LES
Fuel conduction	Sub-scale pin model	3-D finite element (MOOSE)
Transient	Yes (+0.5 \$ insertion)	Out of scope
Solution error	Discretisation, not quantified	Statistical, quantified ($\pm$ pcm)

1.4 Acceptance criterion

The project sets **500 pcm on** k as the flag for a material difference. Section 4 applies this criterion only between methodologically comparable quantities.

2 Results: GeN-Foam

2.1 Solver acceptance

Prior to the TRIGA calculation, GeN-Foam was verified against its own reference cases.

Table 3. GeN-Foam solver acceptance (`validation_step2.csv`).

Case	Quantity	Simulated	Expected	Status
3D SmallESFR	k_{eff}	0.936873	0.936827	PASS (4.9×10^{-5})
2D point kinetics	steady state	—	—	PASS
2D point kinetics	transient, no driveline	—	—	PASS
2D point kinetics	transient, driveline	—	—	PASS
2D point kinetics	transient, boron	—	—	PASS

2.2 Lattice constants

GeN-Foam condensed its own two-group constants from an OpenMC pin-cell calculation (70 batches, 20 inactive, 8000 particles per batch), giving $k_\infty = 1.38810 \pm 0.00123$.

Table 4. GeN-Foam two-group lattice constants (`validation_step5.csv`).

Quantity	Fast	Thermal
Removal (m^{-1})	4.77341	8.49190
$\nu\Sigma_f$ (m^{-1})	0.379899	12.5219
Diffusion coefficient (m)	0.00749023	0.00158363
χ	1.0	0.0
Inverse velocity ($\text{s}\,\text{m}^{-1}$)	6.62997×10^{-6}	3.31676×10^{-4}
$S_{1\to1}$	39.7524	
$S_{1\to2}$	4.22693	
$S_{2\to1}$	0.023426	
$S_{2\to2}$		201.932

CONSISTENCY OF THE GROUP CONSTANTS

Solving the two-group constants above analytically for an infinite medium gives $k_\infty = 1.38873$, against the Monte Carlo lattice value of 1.38810 — a difference of +33 pcm. The condensation therefore preserves the lattice eigenvalue, and the constants are not a significant error source in the core comparison. (The same file also lists `genfoam_smoke_keff` = 1.331; this is a diffusion solution on a separate 2-D channel test geometry with leakage, not on the pin cell, so it is not a method benchmark.)

2.3 Steady state

The core is modelled as a bare homogenised **2-D slab**, 0.495 m wide and 0.60 m tall, one cell thick with `empty` front and back patches (so infinite in the third direction), on a 20×24 mesh with zero flux imposed on the boundary.

Table 5. GeN-Foam steady-state results (`validation_step6.csv`).

Quantity	Value	Note
k_{eff}	1.240544	eigenvalue diffusion, bare homogenised 2-D slab, zero-flux boundary
Power	250 kW	target
Max coolant temperature	306.47 K	fluid T
Max average fuel temperature	594.2 K	sub-scale pin average; well below the 1 023 K limit
Power peaking (max/avg)	2.45	cosine–cosine shape; analytic slab value $\pi^2/4 = 2.47$

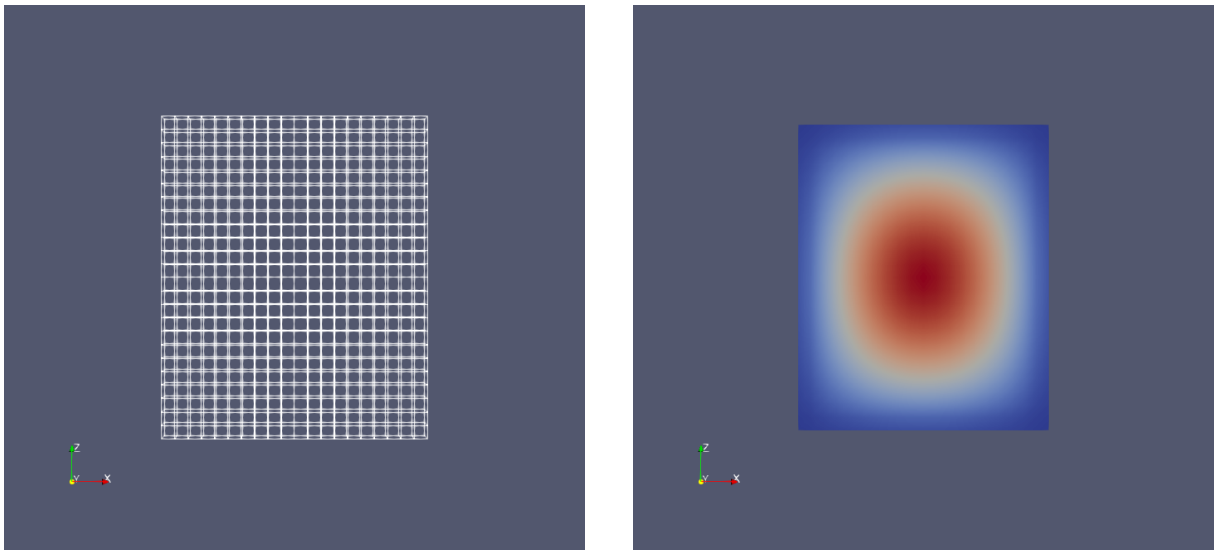


Figure 1. GeN-Foam computational mesh (left) and steady-state power density (right).

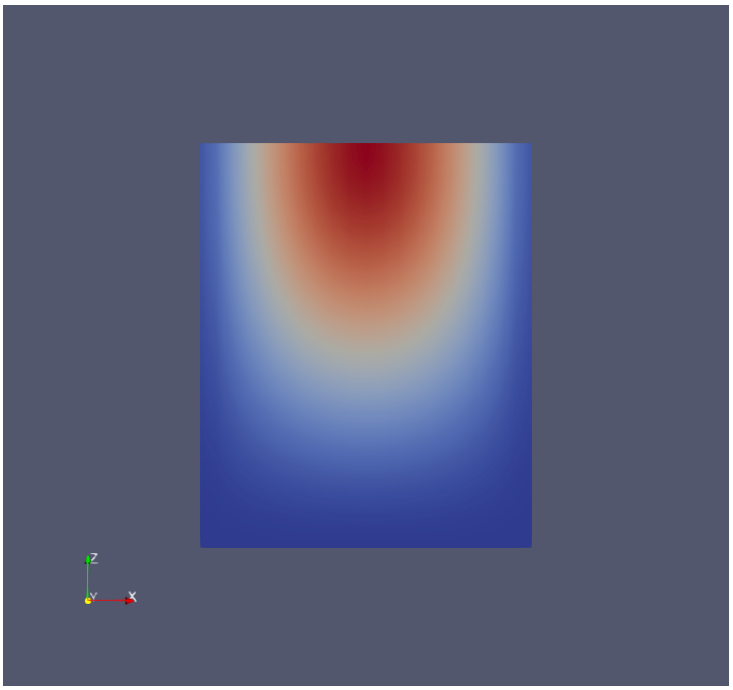


Figure 2. GeN-Foam steady-state temperature field. The maximum lies in the upper portion of the core, consistent with coolant heating as it rises through the channel.

2.4 Reactivity insertion transient

A $+0.5\,\$$ ramp ($0.00325\,\Delta k/k$) was inserted between $t = 100$ and $t = 100.5\,\mathrm{s}$.

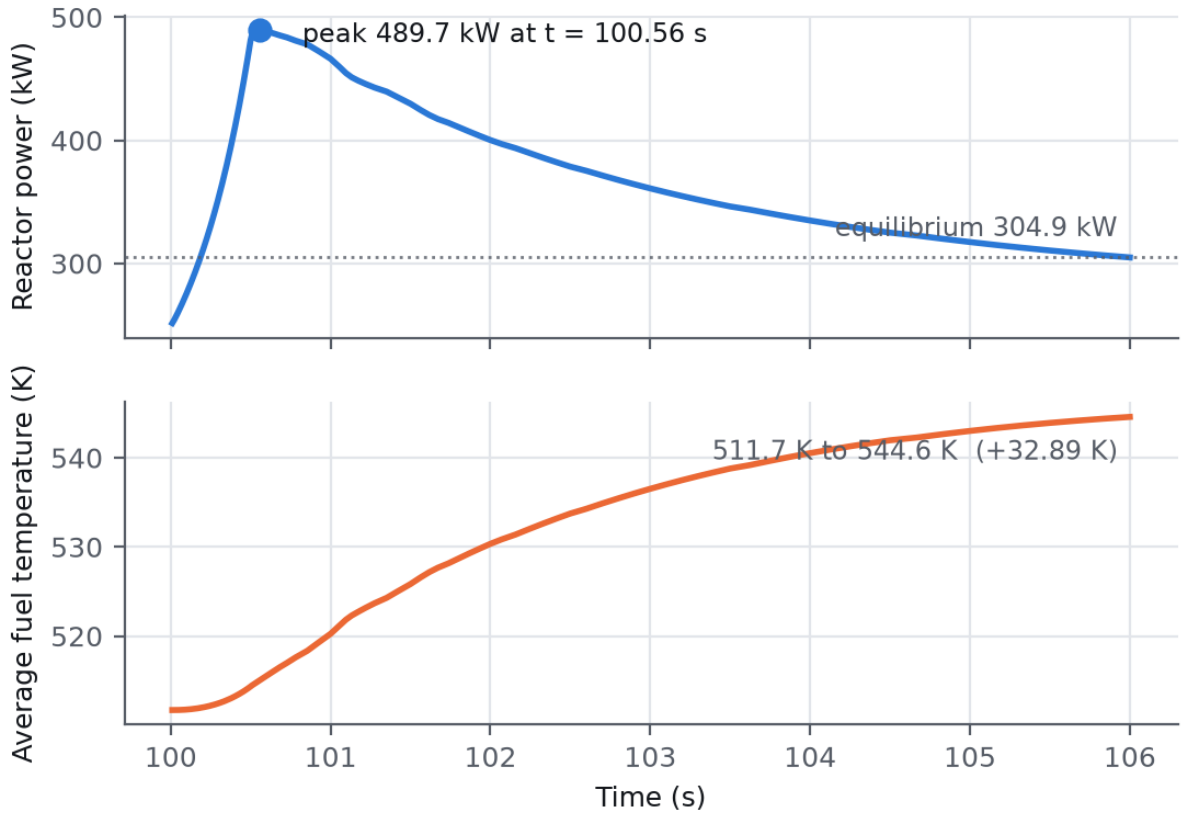


Figure 3. GeN-Foam transient response. Power peaks at 489.7 kW and settles to a Doppler-limited equilibrium of 304.9 kW; average fuel temperature rises 32.89 K.

Table 6. GeN-Foam transient summary.

Quantity	Value	Note
Peak power	489 723 W	at $t = 100.56$ s
Final power	304 884 W	Doppler-limited equilibrium
Final average fuel T	544.59 K	at $t = 106$ s
Net reactivity	−3.87 pcm	external +325 vs Doppler −328.9

FEEDBACK COEFFICIENT IS AN INPUT, NOT A RESULT

The stated Doppler feedback of −328.9 pcm accompanies a fuel temperature rise from 511.70 to 544.59 K, a span of 32.89 K:

$$\frac{-328.9\,pcm}{32.89\,\mathrm{K}} = -10.000\ \mathrm{pcm/K}.$$

The result is *exactly* −10.000 because the coefficient is a constant model input: `feedbackCoeffTFuel` = $-1.0\times10^{-4}\,\mathrm{K}^{-1}$ in the point-kinetics `nuclearData` of the transient case, taken from a published TRIGA prompt fuel temperature coefficient (University of Texas TRIGA overview) rather than computed for this model. The transient uses U-235 Keepin six-group delayed-neutron data and a prompt generation time of 43 µs. Section 4 compares the coefficient against Cardinal's computed value.

3 Results: Cardinal / OpenMC / MOOSE / NekRS

3.1 Method

Continuous-energy Monte Carlo neutronics (OpenMC) is coupled to finite-element heat conduction (MOOSE) through Cardinal by Picard iteration: cell-averaged fission power is transferred to the solid, and cell temperatures are returned to the neutronics. The coolant is treated separately by spectral-element large-eddy simulation (NekRS).

Because the available hardware could not support a full-core coupled model, the coupled calculation uses a **single-pin unit cell**. All coupled temperatures therefore describe an *average* pin.

3.2 Neutronics

Table 7. Cardinal eigenvalue results, ENDF/B-VIII.0, continuous-energy Monte Carlo.

Quantity	Model	Value	Uncertainty
k_∞	2-D unit cell, reflective	1.384207	± 21 pcm
k_{eff}	3-D bare core, heterogeneous	1.102059	± 26 pcm
k_{eff}	3-D bare core, homogenised	1.145259	± 74 pcm
Leakage fraction	bare core	0.20149	± 0.00012
Power peaking (max/avg)	120 pins	2.2544	—
Peak power density	bare core	$6.595\,\mathrm{W\,cm^{-3}}$	—

The homogenised variant was computed specifically to separate geometry treatment from solution method in the comparison of Section 4. Its sensitivity to the $S(\alpha,\beta)$ choice is itself notable: 1.145259 (H in H_2O), 1.132505 (H in ZrH) and 1.136651 (none) span 1275 pcm, more than twice the acceptance criterion.

3.3 Fuel temperature coefficient

The defining TRIGA parameter was computed from first principles by sweeping fuel temperature with the moderator held at 293 K, isolating fuel feedback.

The sweep was performed on the **reflective unit cell**, so the eigenvalue it returns is k_∞ by construction: with reflective boundaries on all six faces there is no leakage, and no k_{eff} exists for that model. This isolates the fuel feedback mechanism itself — the change in the neutron balance within the lattice — from the leakage response of a finite core. Section 3.4 shows why that distinction is quantitatively important here.

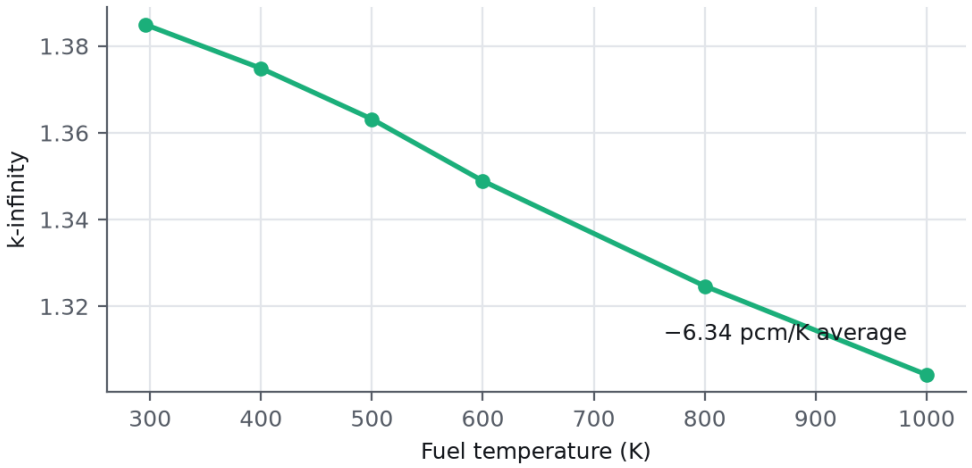


Figure 4. Cardinal fuel temperature coefficient. k_∞ falls from 1.384926 at 296 K to 1.304249 at 1 000 K.

Table 8. Cardinal local fuel temperature coefficient by interval ($\pm 1\sigma$, Monte Carlo statistics).

Band	Coefficient
296–400 K	-5.08 ± 0.43 pcm/K
400–500 K	-6.24 ± 0.46 pcm/K
500–600 K	$\mathbf{-7.76 \pm 0.47}$ **pcm/K**
600–800 K	-6.78 ± 0.23 pcm/K
800–1 000 K	-5.92 ± 0.24 pcm/K
296–1 000 K average	-6.34 pcm/K

The coefficient is large, negative, and becomes *more* negative as the fuel heats from room temperature. Both features identify this as hydride physics rather than U-238 Doppler: a UO_2 LWR pin gives roughly -2 to -3 pcm/K and its magnitude does not grow in this way. The highlighted band is the one in which the GeN-Foam transient operates.

3.4 Leakage feedback: the coefficient on a finite core

The coefficient above is an infinite-lattice quantity. In a finite core the eigenvalue is $k_{\text{eff}} = k_\infty P_{\text{NL}}$, where P_{NL} is the non-leakage probability, so the measurable reactivity coefficient carries a second term:

$$\frac{1}{k_{\text{eff}}}\frac{\partial k_{\text{eff}}}{\partial T_f} = \underbrace{\frac{1}{k_\infty}\frac{\partial k_\infty}{\partial T_f}}_{\text{lattice feedback}} + \underbrace{\frac{1}{P_{\text{NL}}}\frac{\partial P_{\text{NL}}}{\partial T_f}}_{\text{leakage feedback}} .$$

The leakage term is not negligible in this problem, because the bare core leaks **20.1%** of its neutrons. Physically, heating the hydride up-scatters thermal neutrons off the $\sim$0.14 eV ZrH oscillator, hardening the in-fuel spectrum. A harder spectrum has a longer migration length, so a larger fraction escapes a finite core: P_{NL} falls with fuel temperature and the leakage term is *negative*, reinforcing the lattice feedback.

The sweep was therefore repeated on the bare heterogeneous core with vacuum boundaries.

Table 9. Fuel temperature coefficient with and without leakage. The leakage term contributes 20–36% of the total and grows with temperature.

Band (K)	k_∞ (unit cell)	k_{eff} (bare core)	Leakage term	Share
296–400	−5.08	-6.31 ± 0.73	−1.23	20%
400–500	−6.24	-9.12 ± 0.78	−2.88	32%
500–600	−7.76	$\mathbf{-10.79 \pm 0.82}$	−3.04	28%
600–800	−6.78	-10.61 ± 0.42	−3.83	36%
800–1000	−5.92	-9.10 ± 0.44	−3.18	35%

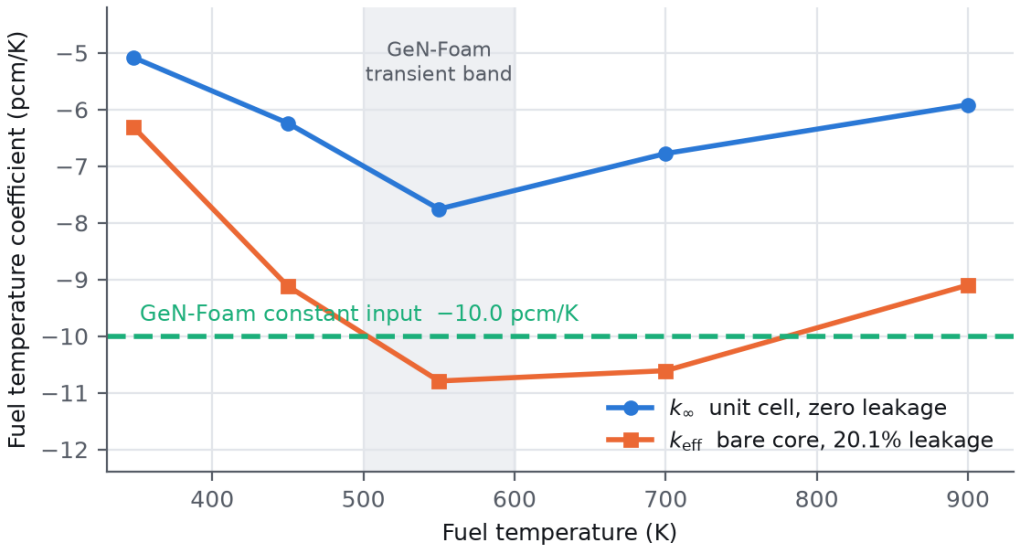


Figure 5. Fuel temperature coefficient computed with and without leakage. All values in pcm/K, plotted at band midpoints.

Both curves suggest a maximum magnitude between about 500 and 800 K, consistent with the up-scattering effect saturating once the thermal population has been pushed clear of the ZrH oscillator energy; the turnover is resolved at roughly the 2σ level only. **The k_{eff}-based coefficient is the one that governs a core transient**, and it is 25–55% more negative than the lattice value across the whole range.

3.5 Coupled neutronics–conduction

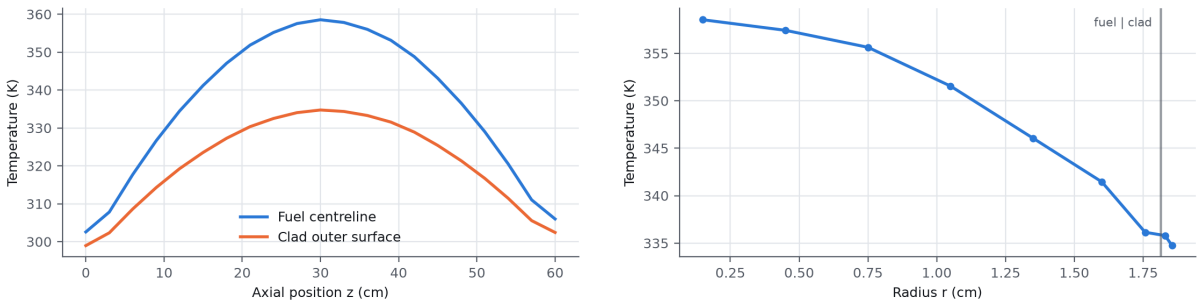


Figure 6. Cardinal coupled temperature solution. Left: axial profile, slightly top-skewed (305.98 K outlet against 302.56 K inlet) because the coolant heats as it rises. Right: radial profile at mid-plane; temperature is continuous across the fuel–clad interface at 1.815 cm, confirming the stitched contact.

Table 10. Cardinal coupled results and temperature drop budget.

Quantity	Value	Analytic	Share
Peak fuel temperature	359.0 K	362.8 K	—
Peak clad temperature	336.2 K	—	—
Power integral	2 083.33 W	2 083.33 W	exact
Film drop (clad → coolant)	40.3 K	42.4 K	63%
Fuel conduction drop	22.4 K	24.1 K	35%
Clad conduction drop	1.07 K	1.5 K	2%

Agreement with the analytic cylindrical-conduction solution is within 1.1%; the residual is explained by the true axial power shape being flatter than the cosine the hand calculation assumes. The film drop dominates and is the least certain term — it derives from a Dittus–Boelter correlation at Re = 4755, which is transitional and below the range where that correlation is reliable.

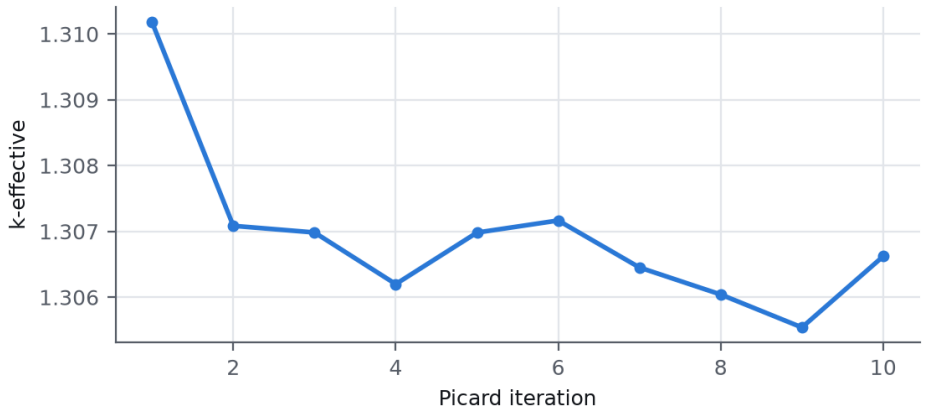


Figure 7. Cardinal Picard convergence. k_{eff} settles within Monte Carlo noise after the first temperature feedback; average fuel temperature is stable to 10^{-8} K.

3.6 Thermal-hydraulics

An implicit large-eddy simulation of the coolant subchannel was run at Re = 4755, polynomial order 7, 2880 spectral elements, on an NVIDIA GTX 1080 Ti. The run completed 20 000 steps but covered only **1.17 flow-through times**. A conservation check after the run also found that only 59% of the clad surface carried the heated-wall boundary tag in the committed mesh; the mesh generator has been corrected but the mesh has not yet been regenerated. The CFD is therefore a **diagnostic of the GPU workflow**, and none of its temperatures is used in a result or comparison.

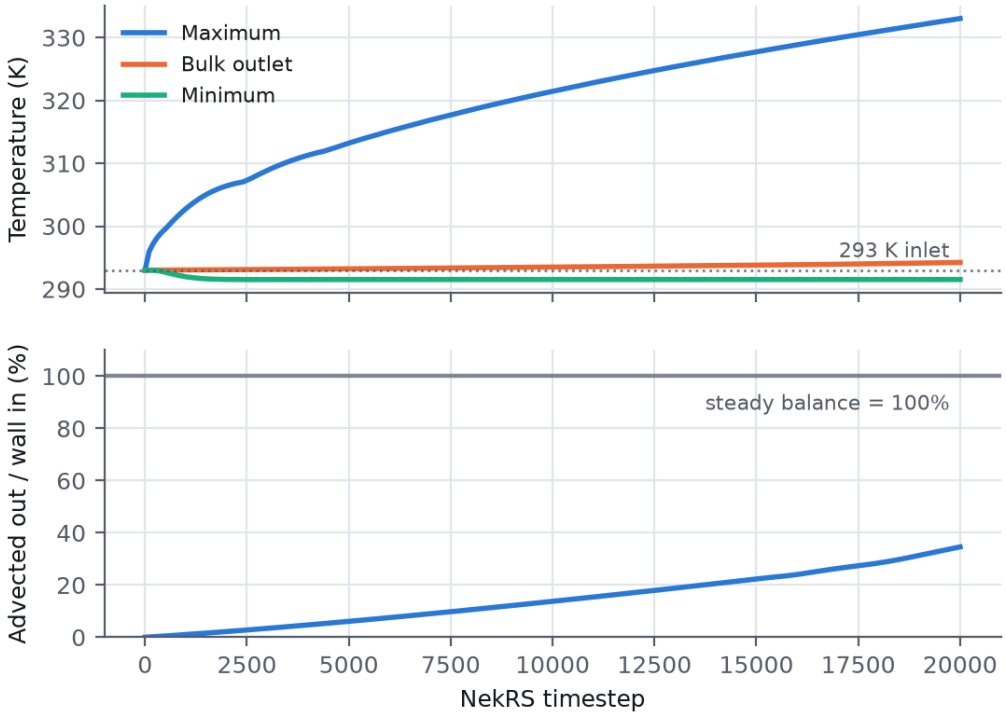


Figure 8. Cardinal NekRS evolution. Upper: coolant temperatures, all still rising at the final step, with the minimum below the 293 K inlet throughout. Lower: energy closure reaches only 34.6% of the steady balance.

Table 11. Cardinal thermal-hydraulic state at 20 000 steps.

Quantity	Value	Steady target	Status
Bulk outlet temperature	294.24 K	296.57 K	not physical
Maximum temperature	333.03 K	—	still rising
Minimum temperature	291.52 K	$\geq$ 293 K	unphysical
Advected heat / wall heat	34.6%	100%	not closed
Mass-flow imbalance	0.122%	$\to$ 0	converged

THE CARDINAL COOLANT CFD IS A DIAGNOSTIC, NOT A RESULT

The flow field has converged — mass imbalance is flat at 0.122% from step 1400 onward — but only 34.6% of the wall heat leaves the domain, so 65.4% is still accumulating. Together with the incomplete heated-wall tagging, this means **the coolant temperatures above are not physical results.** Reaching approximately five flow-throughs would require about 85 700 steps and 5.1 h of GPU time.

A second defect is recorded rather than smoothed over: the minimum temperature lies below the inlet in a domain whose only thermal boundary conditions are a 293 K inlet and a heated wall. With no heat sink this cannot be physical. It is an undershoot from the high-pass filter stabilisation and it does not decay.

3.7 Nuclear data sensitivity

The specification requires ENDF/B-VIII.0. The execution environment for the coupled run carried ENDF/B-VII.1, so the identical coupled model was run on both libraries and the difference recorded as a measured offset rather than left as an unknown.

Table 12. Library sensitivity of the coupled Cardinal model.

Library	**Platform**	k_{eff}	**Max fuel** T	**Tally error**
ENDF/B-VIII.0	macOS arm64, 5k particles	1.30561	358.64 K	2.06%
ENDF/B-VII.1	Ubuntu x86, 20k particles	1.30656	358.90 K	0.82%
Difference	VII.1 − VIII.0	+73 pcm	+0.25 K	—

73 pcm is small relative to the acceptance criterion and within combined counting statistics. The same comparison also demonstrates **cross-platform reproducibility**: average fuel temperature agrees to 7×10^{-4} K between macOS arm64 and Ubuntu x86, across different libraries and different particle budgets.

4 Comparison

4.1 Overview

Table 13. Case-by-case comparison. Only cases marked *yes* support a direct numerical difference; the acceptance criterion applies to those alone.

#	Case	GeN-Foam	Cardinal	Difference	Comparable
1	k_∞, unit cell	1.388100	1.384207	−280 pcm	yes
2	k_{eff}, as modelled	1.240544	1.102059	$\Delta\rho = +10129$ pcm	no[c]
3	k_{eff}, same cylinder	1.140929[a]	1.145259	$\Delta\rho = -331$ pcm	yes
4	Group constants, k_∞	1.38873[a]	—	+33 pcm[b]	internal
5	Fuel temp. coefficient	−10.000	-10.79 ± 0.82	+7%	yes
6	Power peaking	2.45	2.2544	−8.0%	no[c]
7	Max fuel temperature	594.2 K	359.0 K	−235 K	no
8	Max coolant temperature	306.5 K	diagnostic only	—	no
9	Total power	250 kW	250 kW	0	yes
10	Transient peak power	489.7 kW	not run	—	no
11	Two-group MGXS	full set	not generated	—	no
12	Solver acceptance	PASS	not recorded	—	no

[a] GeN-Foam two-group constants solved analytically (`keff_decomposition.py`). [b] Against GeN-Foam's own OpenMC lattice, 1.38810. [c] Different core geometry: 2-D slab vs 3-D cylinder.

4.2 Case 1 — the directly comparable quantity

Both sides solved the same 2-D reflective pin cell with OpenMC continuous-energy Monte Carlo and ENDF/B-VIII.0. This is the only case in which the two calculations solve the same equations on the same geometry.

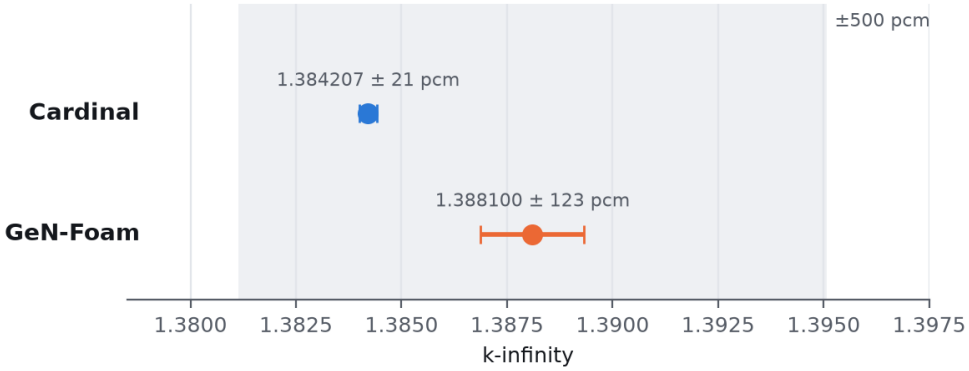


Figure 9. k_∞ comparison. Bars are $\pm 1\sigma$ counting statistics; the shaded band is the 500 pcm acceptance criterion centred on GeN-Foam.

The difference is **−280 pcm** at 3.1σ — statistically resolved but comfortably inside the acceptance criterion. Cardinal's uncertainty is approximately six times tighter, reflecting a 25-fold larger particle budget.

Interpretation. Two separate implementations of the same specification, in the same code, agree. A small part of the residual is expected from a documented input difference: the SS304 cladding was defined at $8.00\,\text{g cm}^{-3}$ in Cardinal and $7.9\,\text{g cm}^{-3}$ with slightly different trace elements in the GeN-Foam lattice. This is the strongest evidence in the study that the shared problem definition was interpreted consistently on both sides.

4.3 Cases 2–4 — the k_{eff} difference is mostly geometry

The two core eigenvalues differ by $\Delta\rho = +10\,129$ pcm, but the two models differ in more than solution method. GeN-Foam's mesh is a 2-D slab (49.5 cm × 60 cm, `empty` front and back, zero flux on the boundary); Cardinal's core is a 3-D cylinder of diameter 49.5 cm and height 60 cm. The slab has no leakage in its third direction, so its geometric buckling is only 56% of the cylinder's.

To separate the effects, GeN-Foam's own two-group constants were solved analytically for a bare homogeneous core in each geometry, and the difference was decomposed into additive reactivity steps ($\rho = 1 - 1/k$). The analysis is reproducible with `keff_decomposition.py` in `results/comparison/`.

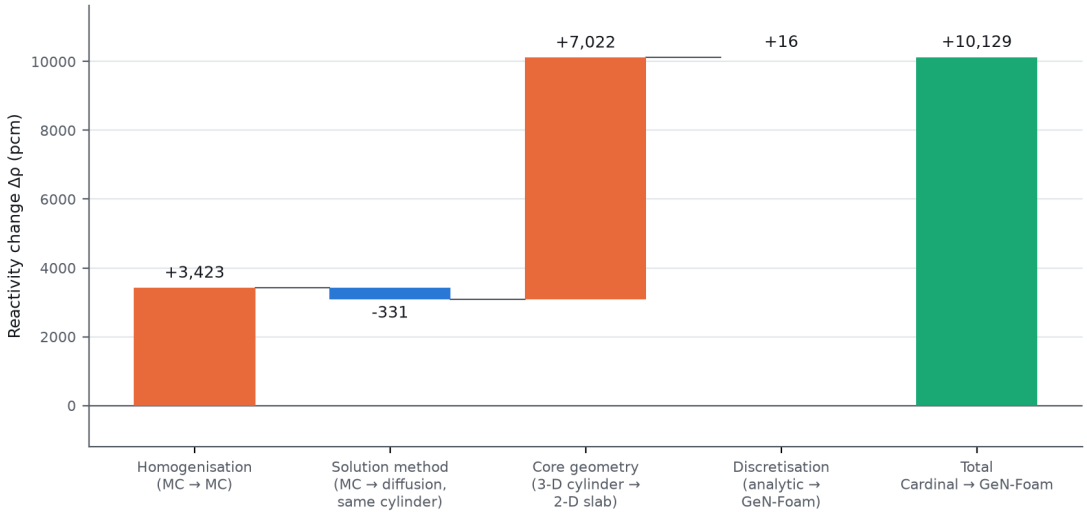


Figure 10. Decomposition of the core eigenvalue difference into additive reactivity steps.

Table 14. Additive decomposition of the core eigenvalue difference.

Step	k: from → to	$\Delta\rho$	What it isolates
1. Homogenisation	1.10206 → 1.14526	+3423 pcm	Monte Carlo both sides; heterogeneous lattice → smeared zone
2. Solution method	1.14526 → 1.14093	−331 pcm	Same homogenised cylinder; Monte Carlo → 2-group diffusion
3. Core geometry	1.14093 → 1.24030	+7022 pcm	Same diffusion constants; 3-D cylinder → 2-D slab
4. Discretisation	1.24030 → 1.24054	+16 pcm	Analytic slab → GeN-Foam finite-volume solution
Total	1.10206 → 1.24054	+10 129 pcm	Equal to the direct difference

Three results follow.

1. **GeN-Foam's solver is verified.** Its numerical eigenvalue reproduces the analytic two-group solution for its own geometry to 16 pcm.
2. **Geometry interpretation dominates.** Reading the core as a 2-D slab rather than a 3-D cylinder accounts for about 70% of the total difference.
3. **With geometry matched, diffusion and Monte Carlo agree closely.** In the same cylinder, diffusion with GeN-Foam's constants and Cardinal's homogenised Monte Carlo

differ by −331 pcm, inside the 500 pcm criterion. With an extrapolated (Marshak) vacuum boundary instead of zero flux the difference is +1468 pcm; both lie within the 1275 pcm spread of the homogenised Monte Carlo result across hydrogen scattering kernels.

The remaining +3423 pcm homogenisation step is the classical two-step error: smearing the lattice removes resonance self-shielding and forces a single hydrogen thermal-scattering kernel on a mixture whose hydrogen is 56% ZrH-bound and 44% water-bound.

CORRECTION TO AN EARLIER VERSION OF THIS REPORT

An earlier version attributed +8320 pcm to diffusion-versus-Monte-Carlo method error, assumed both cores were the same bare box, and treated `genfoam_smoke_keff` = 1.331 as a pin-cell method benchmark. Inspection of the GeN-Foam mesh showed the core is a 2-D slab, and the smoke test was run on a separate channel geometry. The decomposition above replaces that analysis.

4.4 Case 5 — fuel temperature coefficient

This is the most consequential physics quantity, because the coefficient governs the entire transient response.

Table 15. Fuel temperature coefficient in GeN-Foam's operating band (500–600 K). The comparison depends on which eigenvalue the coefficient is referred to.

	Value	**vs GeN-Foam**
Cardinal, k_∞ — unit cell, zero leakage	−7.76 pcm/K	−29%
Cardinal, k_{eff} — bare core, 20.1% leakage	$\mathbf{-10.79 \pm 0.82\ pcm/K}$	**+7%**
GeN-Foam, transient (published constant)	−10.000 pcm/K	—

A transient is driven by the *core* eigenvalue, so the k_{eff}-based coefficient is the correct basis for comparison. On that basis the two values agree to **7%**, within one standard deviation of the Monte Carlo estimate. Because GeN-Foam's slab leaks less than the specified cylinder, the coefficient appropriate to its own geometry would lie between the lattice and cylinder values; the constant input is therefore, if anything, slightly too negative for the slab. The apparent 29% discrepancy against the lattice value is an artifact of comparing an infinite-lattice quantity with a core quantity; see Section 3.4.

WHAT REMAINS OPEN

The magnitude of GeN-Foam's coefficient is *corroborated* by the Cardinal core calculation: −10.0 against −10.79 pcm/K is agreement at the 7% level, so the transient results do not appear to be biased by the choice of coefficient.

Two points nevertheless remain. First, the coefficient is entered as a **constant**, whereas the computed value varies from −6.3 to −10.8 pcm/K over 296–1 000 K. A constant chosen to suit the 500–600 K band is too negative below 450 K and again above 800 K, so a larger excursion reaching higher fuel temperatures would be shut down more strongly than the physics supports. Second, the input is a published value for a reflected TRIGA rather than one computed for this bare-core model, so close agreement is partly fortuitous.

Recommended action: replace the constant with Cardinal's temperature-dependent coefficient for transients leaving the 500–600 K band.

4.5 Cases 6–8 — thermal quantities

These cannot be differenced as reported, for three independent reasons:

1. **Cardinal describes an average pin.** Its own peaking factor of 2.2544 implies a hot channel is materially hotter, so the two fuel temperatures do not describe the same object.

2. **Cardinal's film coefficient is a correlation, not a solution.** Dittus–Boelter at Re = 4755 supplies 63% of Cardinal's total temperature rise, and is applied outside its reliable range.
3. **Cardinal's coolant CFD is a diagnostic only** (34.6% energy closure, incomplete heated-wall tagging), so no Cardinal coolant temperature is available for comparison.

4.6 Cases 10–12 — no Cardinal counterpart

The Cardinal model was scoped to steady state, so it has no transient counterpart. This is a documented scope decision rather than an omission, but it means GeN-Foam's transient results — including the peak power on which the safety argument rests — have **no independent check**. Cardinal two-group constants were not generated either; GeN-Foam condensed its own from OpenMC, and Case 4 shows these preserve the lattice eigenvalue to 33 pcm.

5 Verification status

THIS WORK IS VERIFIED, NOT VALIDATED

Verification asks whether the equations are solved correctly; validation asks whether the equations describe reality, and is establishable only against experimental data. **No experimental data was used in either calculation**, so no part of this study is validated. Agreement between two codes is not a substitute — two codes can agree and both be wrong, particularly when they share a specification, a nuclear data library, or a simplification. Here both share a bare-core model with no reflector and no control rods, so **neither result describes a physical TRIGA at criticality**.

Table 16. Verification status by level.

Level	Status	Evidence
1. Software	MET	Both codes run to completion; GeN-Foam passes its reference cases; Cardinal conserves power exactly at 2 083.33 W
2. Numerical	PARTIAL	Cardinal neutronics converged (particles, seeds, Shannon entropy) and conduction converged; GeN-Foam matches the analytic two-group solution to 16 pcm; CFD is a diagnostic only; no mesh or order refinement study on either side
3. Physics	PARTIAL	Cardinal conduction within 1.1% of analytic; ZrH feedback confirmed from nuclear data; GeN-Foam's feedback coefficient is a constant literature value, not computed for this model
4. Cross-code	PARTIAL	Lattice k_∞ agrees to 280 pcm; core k_{eff} difference fully decomposed, and agrees to 331 pcm once geometry is matched; fuel temperature coefficient agrees to 7%
5. Experimental	NOT ATTEMPTED	No experimental data exists in this project

Table 17. Actions required to close each gap, in priority order.

	Gap	Action	Cost
1	Core geometry differs	Re-run GeN-Foam on the specified 3-D cylinder	1 h CPU
2	Constant feedback coefficient	Replace the literature constant with Cardinal's temperature-dependent coefficient; re-run the transient	1 h CPU
3	CFD mesh tagging	Regenerate `fluid.re2` with the corrected generator; rerun NekRS	2 h GPU
4	CFD not converged	Extend to ∼5 flow-throughs after resolving the filter undershoot	5 h GPU
5	No mesh independence	Repeat at orders 5 and 8; compare integral quantities	8 h GPU
6	Cardinal library deviation	Re-run coupled production on ENDF/B-VIII.0	1 h CPU
7	Reflector and rods	Add a graphite reflector and control rods to both models	—
8	No validation	Obtain TRIGA experimental data	—

6 Limitations

Table 18. Limitations of the present study, with their consequence for interpretation.

Limitation	Consequence
Both cores are bare	Neither model includes a reflector or control rods, so neither describes a physical TRIGA at criticality. This is a *shared* simplification, not a difference between the calculations.
Core geometry differs	GeN-Foam models a 2-D slab, Cardinal a 3-D cylinder; the effect is quantified (+7022 pcm) but the GeN-Foam core has not been re-run in the specified geometry.
Coolant CFD is a diagnostic	Incomplete heated-wall tagging, 34.6% energy closure and a filter undershoot below the inlet temperature; no CFD temperature is used.
Conduction uses a correlation	The Dittus–Boelter film coefficient supplies 63% of the Cardinal temperature rise and is applied at Re = 4755, outside its reliable range.
Average pin, not hot channel	Cardinal's peaking factor of 2.2544 implies a hot channel is materially hotter than the reported temperatures.
Thermal properties assumed	k_{fuel}, k_{clad}, the film coefficient and the coolant rise are documented modelling choices, not specification values.
Library deviation	The Cardinal coupled production run used ENDF/B-VII.1 rather than the specified VIII.0; measured offset +73 pcm.
No mesh independence study	Neither side performed mesh or polynomial-order refinement.
Transient unverified	The transient was computed by one code only and has no independent check.

7 Reproduction

Both calculations, all inputs and all result data are in the GitHub repository `https://github.com/kavyawadhwa134/triga-crosscode-verification`.

7.1 GeN-Foam

Path	Contents
`genfoam/lattice/`	OpenMC pin cell and two-group MGXS extraction
`genfoam/tools/openmc2genfoam.py`	MGXS → GeN-Foam `nuclearData` converter
`genfoam/triga_steady/`	Steady-state eigenvalue case (`Allrun`, `Allclean`)
`genfoam/triga_transient_rod/`	+0.5 $ insertion transient (restarts from the steady case)

7.2 Cardinal / OpenMC / MOOSE / NekRS

Step	Command
Environment audit	`./scripts/audit_environment.sh`
Neutronics	`./scripts/run_openmc.sh unitcell production --seed 11`
Coupled	`./scripts/run_coupled.sh production 8`
Thermal-hydraulics	`./scripts/run_nekrs.sh 8`
Temperature coefficient	`python cardinal/openmc/temperature_coefficient.py`
Eigenvalue decomposition	`python results/comparison/keff_decomposition.py`

Table 19. Numerical settings for the Cardinal production runs.

Parameter	Value
Standalone neutronics	40 000 particles, 500 batches, 100 inactive (2×10^7 histories)
Coupled neutronics	20 000 particles, 150 batches, 30 inactive, 10 Picard iterations
Solid mesh	6400 elements, 20 axial layers
CFD mesh	2880 spectral elements, polynomial order 7, 1 474 560 points
Coupled runtime	2 905 s wall, 1 132 MB peak
CFD runtime	4 294 s wall, 2 660 MB device, GTX 1080 Ti

7.3 Result files

Path under `results/`	Contents
`genfoam/validation_step{2,5,6}.csv`	GeN-Foam acceptance, lattice, core, transient
`genfoam/transient_rod_power.csv`	GeN-Foam transient time history
`cardinal/openmc_unitcell.csv`	Cardinal k_∞, all configurations and seeds
`cardinal/openmc_core*.csv`	Cardinal k_{eff}, heterogeneous and homogenised
`cardinal/*temperature_coefficient.csv`	Temperature-coefficient sweeps, lattice and core
`cardinal/coupled_*.csv`	Coupled results per Picard iteration
`cardinal/thermal_profiles.csv`	Axial and radial temperature profiles
`cardinal/nekrs_convergence.csv`	CFD convergence history (diagnostic)
`comparison/crosscode_comparison.csv`	Case-by-case comparison, machine-readable
`comparison/keff_decomposition.csv`	Additive eigenvalue decomposition

8 Conclusions

1. **Where the two calculations solve the same problem, they agree.** The Monte Carlo pin cell differs by 280 pcm at 3.1σ, inside the 500 pcm acceptance criterion.

2. **The core eigenvalue difference is explained in full.** Of the +10 129 pcm difference, +7022 pcm comes from GeN-Foam modelling the core as a 2-D slab rather than the 3-D cylinder used by Cardinal, and +3423 pcm from homogenisation. With geometry matched, two-group diffusion and Monte Carlo agree to 331 pcm (1468 pcm with an extrapolated boundary), and GeN-Foam reproduces the analytic solution of its own geometry to 16 pcm.

3. **The ambiguous core dimension is the most important specification issue.** "0.495 × 0.60 m" was read differently by the two models, and that reading is worth about 7000 pcm. A benchmark specification should state the core shape explicitly.

4. **The fuel temperature coefficient is corroborated, with a caveat.** Cardinal computes -10.79 ± 0.82 pcm/K for the bare core in the 500–600 K band from ZrH scattering data, within 7% of the published -10.0 pcm/K constant used by GeN-Foam. The constant does not capture the computed variation from -6.3 to -10.8 pcm/K over 296–1 000 K.

5. **Incomplete parts are reported as such.** The Cardinal coolant CFD is a diagnostic only, and Cardinal did not attempt the transient. Neither is presented as a result.

6. **Neither calculation is validated.** Both cores are bare, without reflector or control rods, and no experimental data was used. The study establishes cross-code consistency — not fidelity to a physical reactor.

Inputs, results and analysis scripts for both calculations: `https://github.com/kavyawadhwa134/triga-crosscode-verification`. Cardinal results were frozen at tag `pcm-independent-v1` before any GeN-Foam result was read.